Graph-Based Cross-Channel Mule Account Detection System Using Machine Learning

Team Name: Tech Whiz

Team Members:

Member 1 – Anees Fathima B I
Member 2 – Abinaya S
Member 3 – Indhu S
Member 4 – Jasmin Fathima P

Institution: Bannari Amman Institute of Technology, Sathyamangalam, Erode, TamilNadu

Introduction

Financial fraud has increased significantly with the rapid adoption of digital banking, mobile payment platforms, and online financial services.

- Digital transactions generate massive volumes of financial data
- Fraudsters exploit vulnerabilities across multiple banking platforms
- Mule accounts are frequently used to move and disguise illegal funds
- Complex transaction relationships are difficult to detect using traditional monitoring systems
- Advanced analytical techniques are required to uncover hidden financial networks

Mule Accounts and Fraud Networks

Mule accounts are bank accounts used by criminals to transfer illegally obtained funds through multiple intermediaries.

Individuals allow criminals to use their accounts to move fraudulent funds

Transactions are routed through multiple accounts to hide the original source

This creates **complex transaction networks involving several accounts and devices**

Fraud networks often span across banking platforms, digital wallets, and payment gateways

Detecting hidden relationships between accounts is critical for fraud prevention

Limitations of Traditional Fraud Detection Systems

Traditional fraud detection mechanisms rely mainly on rule-based monitoring and isolated transaction analysis.

- Monitoring is usually limited to a **single financial platform**
- Rule-based systems fail to detect sophisticated multi-account fraud patterns
- Cross-platform fraud activities are difficult to correlate
- Large transaction volumes make manual analysis inefficient
- Relationship patterns between accounts and devices remain hidden

Problem Statement and Objectives

Problem Statement

Financial institutions struggle to detect coordinated fraud because criminals distribute transactions across multiple accounts and platforms using mule accounts.

Objectives

- Detect mule accounts involved in suspicious financial transactions
- Model relationships between accounts, devices, and transactions
- Identify abnormal transaction patterns using graph-based analytics
- Apply machine learning models to classify suspicious accounts
- Generate alerts for high-risk accounts and fraudulent transaction networks

Proposed Solution

The proposed system uses **graph analytics combined with machine learning models** to detect mule account networks.

Financial transactions are represented as a **directed graph structure**

Nodes represent entities such as bank accounts, users, devices, or IP addresses

Edges represent transactions or interactions between these entities

Graph structures capture hidden relationships that traditional systems cannot detect

Advanced analytical methods include:

- **Graph traversal algorithms** to track the flow of funds across multiple accounts
- **Centrality measures** to identify highly influential or suspicious nodes
- **Community detection algorithms** to identify clusters of coordinated fraudulent accounts
- **Temporal transaction analysis** to detect rapid or abnormal transaction sequences

Machine learning models analyze these graph-based features to classify accounts as legitimate or suspicious.

Project Workflow

The system follows a structured pipeline integrating data processing, graph modeling, and machine learning.

01

Data Ingestion

Transaction data is collected including account ID, transaction amount, timestamp, device ID, and IP address.

02

Data Preprocessing

Data is cleaned, normalized, and formatted for analysis.

03

Graph Construction

A transaction graph is created where accounts are nodes and transactions are directed edges.

04

Feature Engineering

Graph-based features such as node degree, transaction frequency, clustering coefficient, and connectivity patterns are extracted.

05

Model Training

Machine learning algorithms are trained using these features to detect suspicious behavior.

06

Fraud Detection

Accounts are assigned risk scores and suspicious accounts are flagged for investigation.

System Implementation

The system integrates **data pipelines, graph analytics modules, and machine learning components**.

- Transaction data is stored and processed through backend data pipelines
- Graph structures are generated using libraries such as NetworkX or graph databases such as Neo4j
- Graph queries and traversal operations analyze relationships between accounts and devices

Graph feature extraction includes:

- In-degree and out-degree transaction counts
- PageRank score to identify influential accounts
- Betweenness centrality to detect accounts acting as intermediaries
- Transaction velocity and frequency metrics

Machine Learning Layer

- Extracted graph features are input into classification models such as Random Forest or Gradient Boosting
- Models learn behavioral patterns associated with fraudulent networks
- Accounts exceeding predefined risk thresholds are flagged as mule accounts

Alert Generation

- Suspicious accounts are reported through monitoring dashboards
- Financial analysts can further investigate flagged networks

Technology Stack

Programming and Data Processing

- Python
- Pandas and NumPy

Graph Analytics

- NetworkX
- Neo4j Graph Database

Machine Learning

- Scikit-learn
- TensorFlow or PyTorch

Backend Services

- Node.js
- Express.js APIs

Database Systems

- MongoDB
- PostgreSQL

Visualization

- Power BI or Tableau
- Graph visualization tools

Expected Outcome and Conclusion

The proposed system improves fraud detection by analyzing relationships within financial transaction networks.

Detect mule accounts involved in illegal fund transfers

Identify hidden connections between accounts, devices, and transactions

Improve fraud detection accuracy using graph analytics and machine learning

Enable early alerts for suspicious financial activities

Help financial institutions prevent large-scale financial fraud

This system provides a **scalable and intelligent framework for cross-channel fraud detection in modern banking environments.**